out-competed by more aggressive starlings. Therefore their realized niche—the places they actually occupy—is not as extensive as their potential (or fundamental) niche.

Many animals share some aspects of their niche, but not others. This is called niche overlap. If different species live in the same habitat and have similar lifestyles, they will be in competition but they may be able to live in close proximity if some aspects of their behavior or diet differ. This arrangement is known as niche partitioning. For example, various anole lizards on Puerto Rico successfully occupy the same areas because they select perching locations in different parts of trees.

There are limits to niche overlap. When two animals with identical niches live in the same place, one will drive the other to extinction. This concept—the competitive exclusion principle—was outlined by Joseph Grinnell in 1904 and developed in a paper published by Russian ecologist Georgy Gause in 1934, becoming known as Gause’s law.